Recommendations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Recommendations before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Similarity

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Prediction Is Not Fact

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Recommendations trace and record the first step you can explain without help.

Use W2: Explain a **recommendation** honestly for **Recommendation**s and feedback. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Recommendations explanation against the supplied facts before discussing it.

Use the worked case for **Recommendation**s and feedback. Proposed explanation: “A high score can cancel a failed requirement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the process to activities

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Recommendations target to a trace and a justified conclusion.

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Recommendations case and show the deciding evidence.

A recommender suggests adventure books after one adventure-book choice.

Explain its **evidence** and limitation.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A recommender suggests a space book because you liked two space books. Does the suggestion prove you will enjoy it? What feedback could help next time?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 20 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Recommendations adds to Helpful Classroom Sorter and keep the evidence.

If the sorter ranks bins, state the **objective** and show why the top suggestion is not a command.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A system shows only adventure books and records only adventure clicks. It concludes the reader dislikes poetry. Explain the missing **evidence** and propose a small test.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Filter before scoring

> Filter before scoring. Apply the supplied Recommendations facts and keep your reasoning.

Apply BookRec1 to J, K, L and M from the lesson. State **eligibility** or exclusion reason for each, the scores of eligible books and the **recommendation**. Explain why L's two **preference** matches do not make it the winner.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Explain a recommendation honestly

> Explain a recommendation honestly. Apply the supplied Recommendations facts and keep your reasoning.

Write an explanation of why J was recommended that mentions one **preference** it does not match. A marketer adds guaranteed enjoyment. Explain why that phrase is unsupported.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Do not label an unseen item rejected

> Do not label an unseen item rejected. Apply the supplied Recommendations facts and keep your reasoning.

J was shown and clicked. K was never shown. Separate the observed records from the labels enjoyed and disliked. Write a question that could gather a stated reason while allowing no answer.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the process to activities

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the process to activities. Apply the supplied Recommendations facts and keep your reasoning.

A fictional learner needs an indoor activity lasting at most 20 minutes. **Preference**s score 2 for drawing and 1 for solo.

Candidates: A drawing, solo, outdoor, 10 minutes; B drawing, group, indoor, 20 minutes; C music, solo, indoor, 15 minutes; D drawing, group, indoor, 18 minutes. Filter, score and return all highest scorers.

Only B is then shown and selected; C and D are not shown. State what the feedback establishes and what remains unknown.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Handle an empty eligible set

> Handle an empty eligible set. Apply the supplied Recommendations facts and keep your reasoning.

Return to BookRec1. J and K become unavailable, L remains Spanish, and M's availability becomes unknown. A designer proposes recommending M anyway because it matches both **preference**s.

Apply the requirements and propose an honest response.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

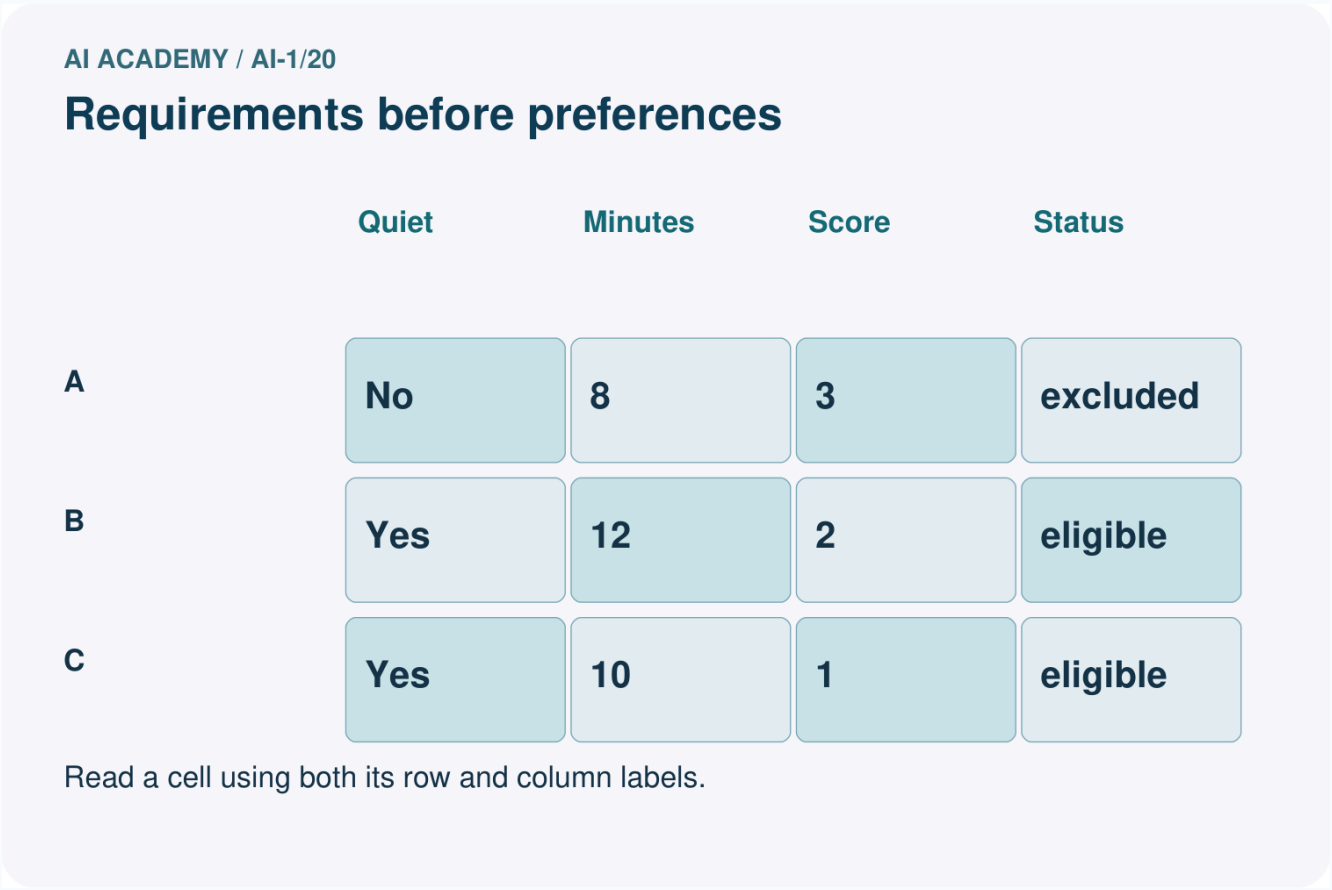
Pictorial case practice: Recommendations

> A fictional classroom break chooser recommends a suitable paper activity. Explain the labelled step before checking the changed condition.

A fictional classroom break chooser recommends a suitable paper activity.

Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/20

Recommendations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Requirements

Reject A because it is noisy despite a preference score of 3.



2. Eligible candidates

B and C both satisfy time and quiet requirements.



3. Preference scores

B scores 2; C scores 1.



4. Recommend

Choose B; preserve the reasons and any later feedback.

CHANGE ONE THING

Change the maximum time to 10 minutes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why does the **recommendation** change without changing anyone's **preference**s?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom break chooser recommends a suitable paper activity.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Choose B; preserve the reasons and any later feedback.

1. Recommend

2. Requirements

3. Eligible candidates

4. **Preference** scores

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Recommend Choose B; preserve the reasons and any later feedback. A hard requirement determines whether an option can be chosen at all.

Preferences rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the maximum time to 10 minutes. Which conclusion is supported by the supplied facts?

1. Only C remains eligible and is recommended.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Only C remains eligible and is

recommended. B takes **12 minutes**; A is still noisy.

Repair a missing building block

Review Similarity (ai-1/19). Goal: Calculate similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

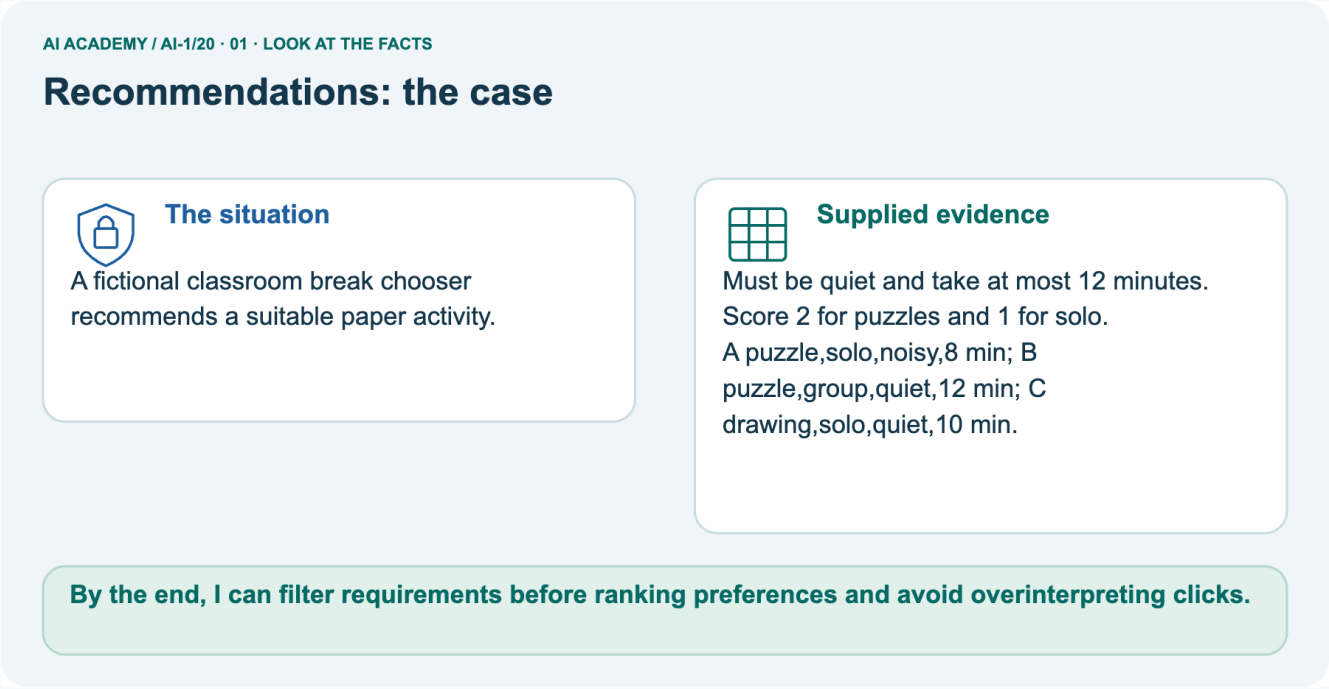
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Recommendations picture from its labels, then record a relevant limit.

01. Recommendations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

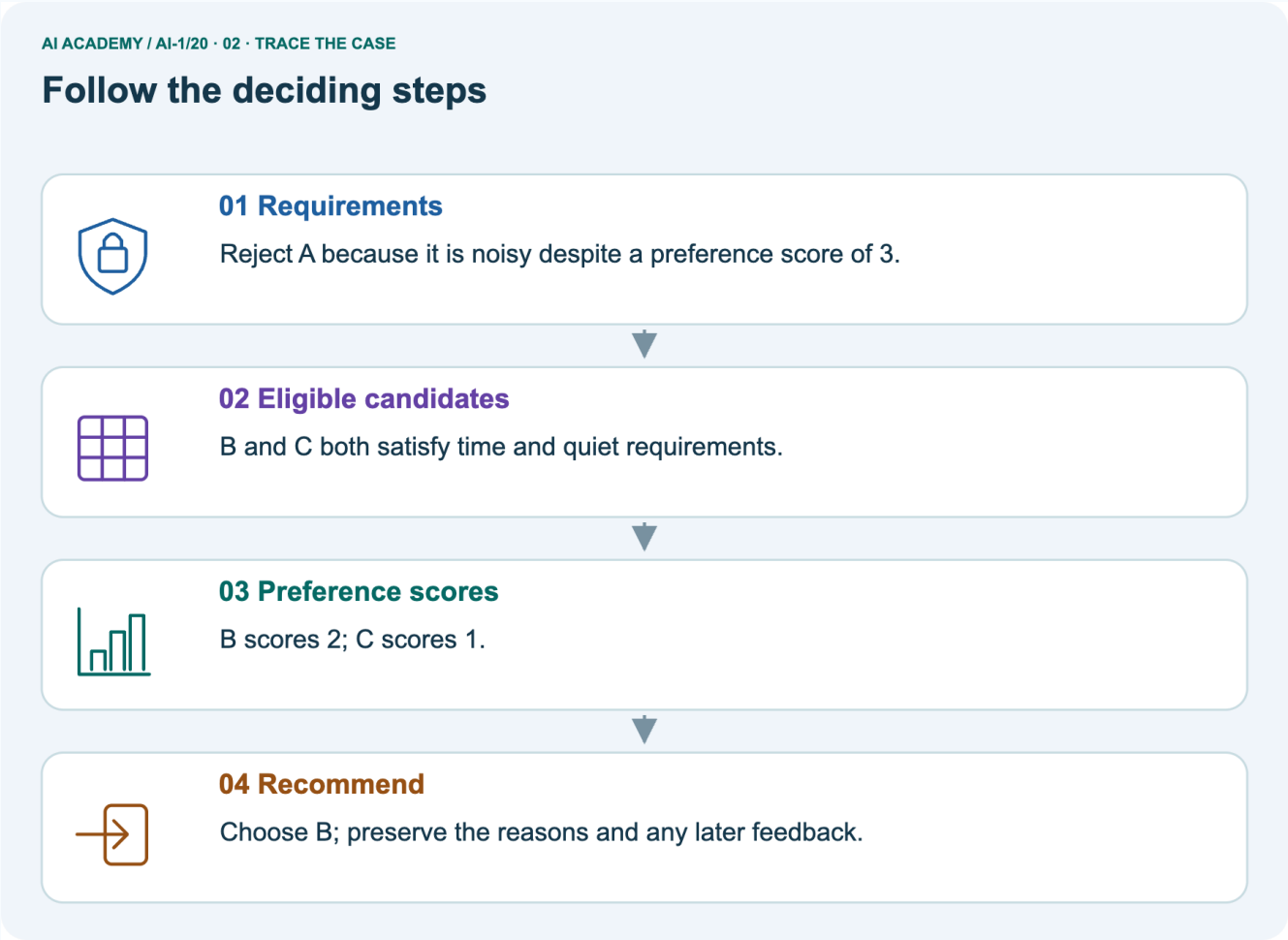
A fictional classroom break chooser recommends a suitable paper activity. Supplied facts: Must be quiet and take at most **12 minutes**. Score 2 for puzzles and 1 for solo.

A puzzle,solo,noisy,8 min; B puzzle,group,quiet,12 min; C drawing,solo,quiet,10 min.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



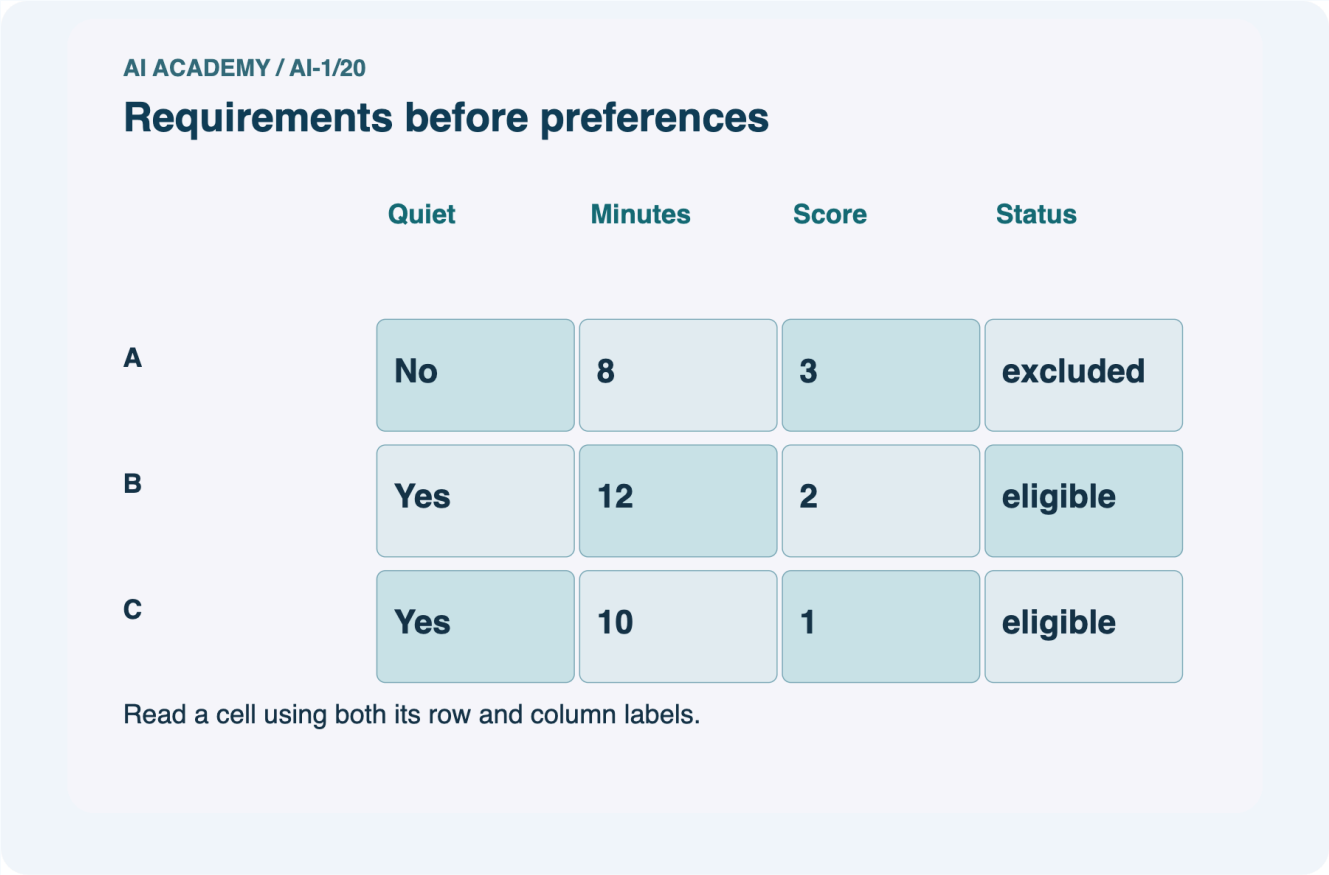
Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements: Reject A because it is noisy despite a **preference** score of 3. → Eligible candidates: B and C both satisfy time and quiet requirements. → **Preference** scores: B scores 2; C scores 1. → Recommend: Choose B; preserve the reasons and any later feedback.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Requirements before preferences



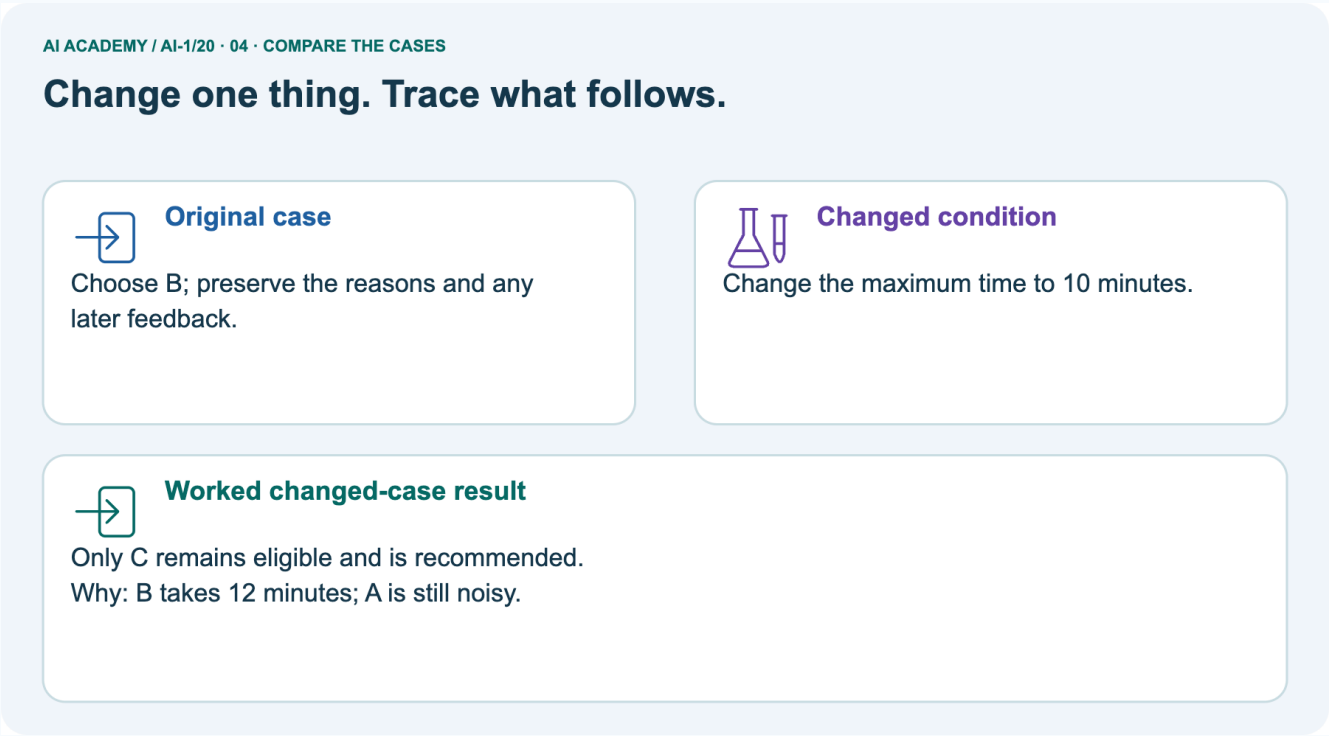
Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements before **preference**s. A: Quiet = No, Minutes = 8, Score = 3, Status = excluded. B: Quiet = Yes, Minutes = 12, Score = 2, Status = eligible. C: Quiet = Yes, Minutes = 10, Score = 1, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

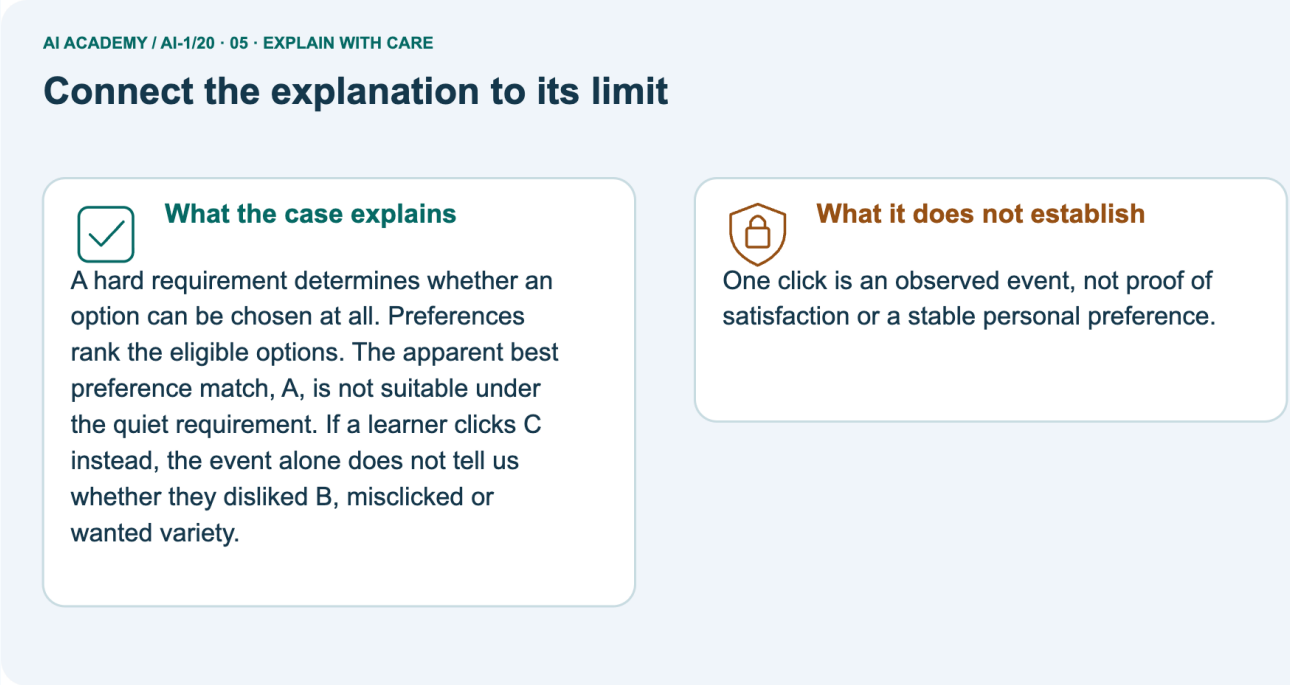
Original case: Choose B; preserve the reasons and any later feedback. Changed condition: Change the maximum time to 10 minutes. Expected result: Only C remains eligible and is recommended.

Reason: B takes **12 minutes**; A is still noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

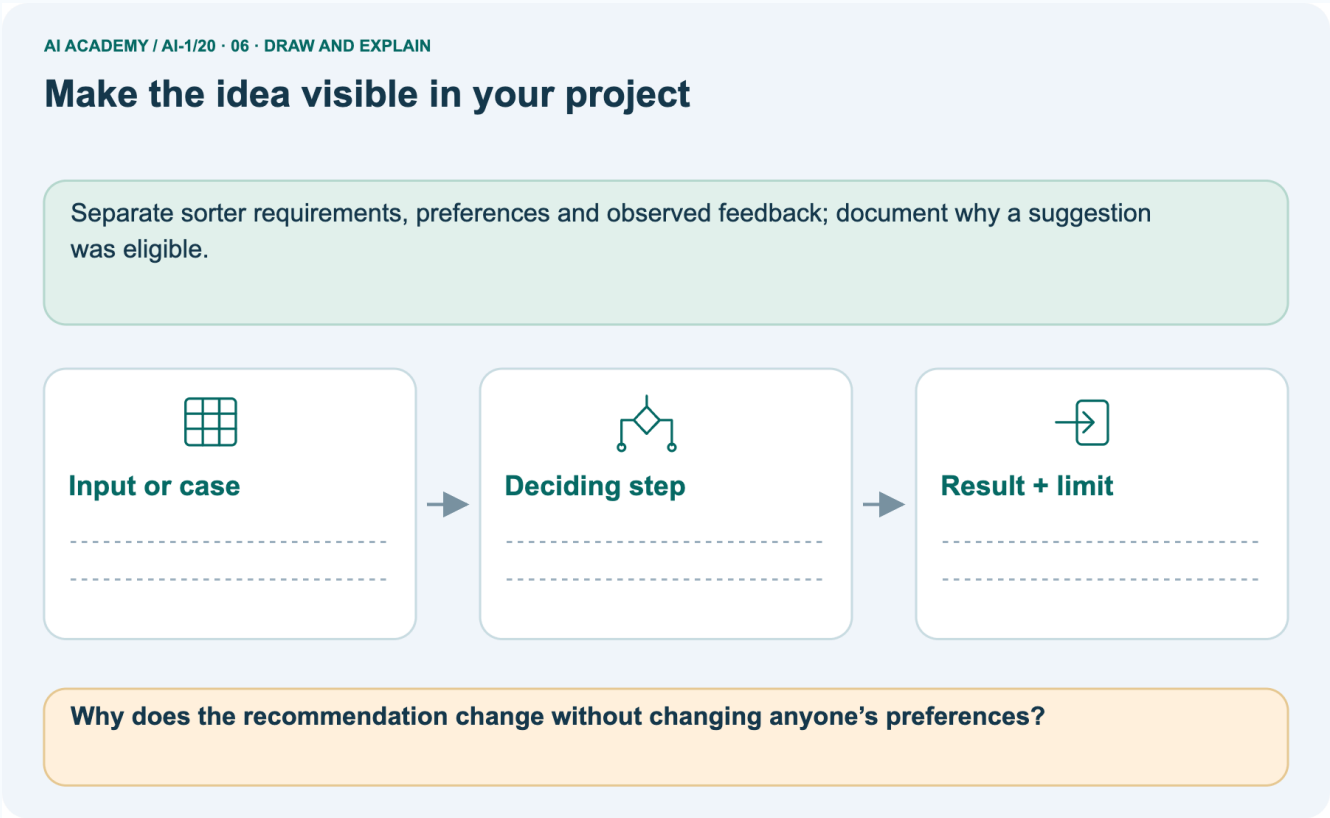
A hard requirement determines whether an option can be chosen at all. **Preference**s rank the eligible options. The apparent best **preference** match, A, is not suitable under the quiet requirement.

If a learner clicks C instead, the event alone does not tell us whether they disliked B, misclicked or wanted variety. Limit: One click is an observed event, not proof of satisfaction or a stable personal **preference**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Separate sorter requirements, **preference**s and observed feedback; document why a suggestion was eligible. Draw your **input**, deciding step, result and limit.

Supported question: Why does the **recommendation** change without changing anyone's **preference**s?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.